

Generation Interconnection System Impact Study Report

For Queue Project Q22-2234

Revere 100 kV
January 30, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2234. The Interconnection Customer proposes a 164.9 MW solar plus storage facility interconnecting at the Revere 100 kV substation in the Indiana area. The request was received on November 21, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 677 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$505,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Segs 7 / Concord 100 kV Circuit 1	Segs 7 / Concord 100 kV Ckt 2	106.9	264	33.6
Freehaven-Lakepark / Segs 7 100 kV #1	Segs 7 / Freehaven-Lakepark 100 kV	106.7	289	33.6
Kemper County / Dalreed 100 kV	Table Mt / Gaston Rd 100 kV	102.3	455	72.6
Gaston Rd / Table Mt 100 kV	Kemper County / Dalreed 100 kV	100.3	529	67.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Segs 7 / Concord 100 kV Circuit 1	Reconductor line	\$1,394,000	15% (\$209,000)
Freehaven-Lakepark / Segs 7 100 kV #1	Reconductor line	\$941,000	15% (\$141,000)
Kemper County / Dalreed 100 kV	Reconductor line	\$376,000	22% (\$84,000)
Table Mt / Gaston Rd 100 kV	Reconductor line	\$330,000	22% (\$71,000)
	Total allocated to project		\$505,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.